

Higgs bundles and minimal surfaces in non-compact symmetric spaces

学习建议

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下面是学习non-Abelian Hodge理论, 极小曲面, Higher Teichmuller 理论的一些学习导引和文献参考。由于这个课程主题比较广泛, 不是一个单一的学习脉络。

1. Non-Abelian Hodge理论。

建议先学习复流形和向量丛上的微分几何, 学习全纯结构, 度量, 联络, 曲率等概念。有两个很好的参考书是

1. Differential Geometry of Complex Vector Bundles, by Shoshichi Kobayashi (前两章即可)
2. D. Huybrechts, Complex geometry : an introduction, Universitext, Springer-Verlag Berlin Heidelberg, 2005. (前4章)

可以接着学习主丛上的度量, 联络和曲率, 有一个很好的短讲义是
Fiber bundles and Chern-Weil Theory, by Johan Dupont

关于调和映照和Teichmuller空间的关系, 可以学习
The Teichmüller theory of harmonic maps, by Michael Wolf

接着就可以学习 Non-Abelian Hodge理论, 学习这方面可以看以下一些笔记或综述:

- [1] Qionglings Li, An introduction to Higgs bundles via harmonic maps, [SIGMA](#) 15 (2019), 035, 30 pages.
- [2] Wentworth R.A., Higgs bundles and local systems on Riemann surfaces, in Geometry and Quantization of Moduli Spaces, Adv. Courses Math. CRM Barcelona, Birkhäuser/Springer, Cham, 2016, 165–219, arXiv:1402.4203.
- [3] Guichard O., An introduction to the differential geometry of flat bundles and of Higgs bundles, in The Geometry, Topology and Physics of Moduli Spaces of Higgs Bundles, Lect. Notes Ser. Inst. Math. Sci. Natl. Univ. Singap., Vol. 36, World Sci. Publ., Hackensack, NJ, 2018, 1–63.

另外可以学习黄鹏飞的一个短期课程 (包括课程视频和讲义) :
Short course on Higgs bundles and local systems
<http://www.cim.nankai.edu.cn/2022/0627/c11453a460256/page.htm>

原始关于Higgs丛和non-Abelian Hodge理论的经典文献是:

- [27] Hitchin N.J., The self-duality equations on a Riemann surface, Proc. London Math. Soc. 55 (1987), 59–126.
- [28] Hitchin N.J., Lie groups and Teichmuller space, Topology 31 (1992), 449–473.

Simpson C.T., Constructing variations of Hodge structure using Yang–Mills theory and applications to uniformization, J. Amer. Math. Soc. 1 (1988), 867–918.

[11] Corlette K., Flat G -bundles with canonical metrics, J. Differential Geom. 28 (1988), 361–382.

[16] Donaldson S.K., Twisted harmonic maps and the self-duality equations, Proc. London Math. Soc. 55 (1987), 127–131.

2. 欧式空间中的极小曲面。

由于这方面很经典，已有很多好教材。

可以先学习极小曲面与单复变之间的关系：Weierstrass 表示和 Gauss map。经典的欧式三维空间中的极小曲面的教材

A survey of minimal surfaces, Robert Osserman

再学习 Second variation of area, Morse index, stability, Simon's identity, Curvature estimate 等内容。

Tobias Colding and William Minicozzi, A Course in Minimal Surfaces. (前两章)

关于欧式三维空间中的极小曲面更多的研究和公开问题，可以参见

A survey on classical minimal surface theory BY William H. Meeks III and Joaquin Perez

关于欧式三维空间中的极小曲面的 Morse index 方面，可以看一个 Notices of AMS 的 survey.

The Morse Index of a Minimal Surface, by Otis Chodosh and Davi Maximo

关于高维欧式空间中的极小曲面，可以读

Complete minimal surfaces in Euclidean space, by S.S. Chern, R. Osserman

经典文章有

1. D. Fischer-Colbrie, On complete minimal surfaces with finite Morse index in three-manifolds, Invent. Math. 82 (1985), no. 1, 121–132, DOI 10.1007/BF01394782. MR808112
2. Doris Fischer-Colbrie and Richard Schoen, The structure of complete stable minimal surfaces in 3-manifolds of nonnegative scalar curvature, Comm. Pure Appl. Math. 33 (1980), no.

3. Higher Teichmüller 理论。

首先学习经典的 Teichmüller 理论，可以学习

A Primer on Mapping Class Groups, by Benson Farb and Dan Margalit

(章节 1, 10, 11)

再学习 Anosov 表示，作为 Higher Teichmüller 理论的关键研究对象。关于 Anosov 表示有一个很好的讲义，几乎可以作为教材，从双曲群讲起，接着介绍 convex cocompact 表示，Anosov 表示的各种等价定义。（可以学 Part 1, 2, 3, 5, 6）由于 Anosov 表示近十年发展迅速，等价定义众多，建议直接读这个讲义，比读多个原始文章要清楚。

Anosov representations: Informal lecture notes, by Richard Canary

接着可以看Anna Wienhard 的一个综述，理解这个领域所关注的方向及问题。

An invitation to higher Teichmüller theory, by Anna Wienhard

同时推荐Francois Labourie在Berkeley的一个关于higher Teichmuller理论的课程讲义：

<https://flab.perso.math.cnrs.fr/Hitchin/index.html>

课程中特别研究orbit map 和well-displacing的文章可以参见

T. Delzant, O. Guichard, F. Labourie, and S. Mozes, “Well displacing representations and orbit maps,” in *Geometry, Rigidity, and Group Actions*, University of Chicago Press, 2011, 494–514.

经典的文章有

1. F. Labourie, “Anosov flows, surface groups and curves in projective space,” *Invent. Math.* 165(2006), 51–114.
2. F. Labourie. “Cross Ratios, Surface Groups, $SL_n(\mathbb{R})$ and Diffeomorphisms of the Circle,” *Publ. Math. de l’I.H.E.S.* 106(2007), 139–213.
3. F. Labourie, “Cross ratios, Anosov representations and the energy functional on Teichmuller space,” “ *Ann. E.N.S.* 41(2008), 439–471.
4. O. Guichard and A. Wienhard, “Anosov representations: Domains of discontinuity and applications,” *Invent. Math.* 190(2012), 357–438.
5. M. Bridgeman, R. Canary, F. Labourie and A. Sambarino, “The pressure metric for Anosov representations,” *G.A.F.A.* 25(2015), 1089–1179.
6. M. Kapovich, B. Leeb and J. Porti, “Anosov subgroups: Dynamical and geometric characterizations,” *Eur. Math. J.* 3(2017), 808–898.